Object Detection Pipeline

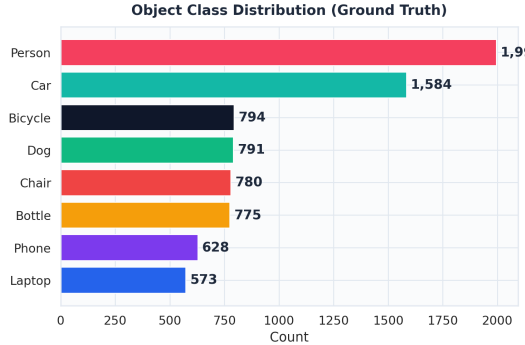
Multi-Model Benchmarking with mAP, Error Taxonomy & Speed-Accuracy Analysis

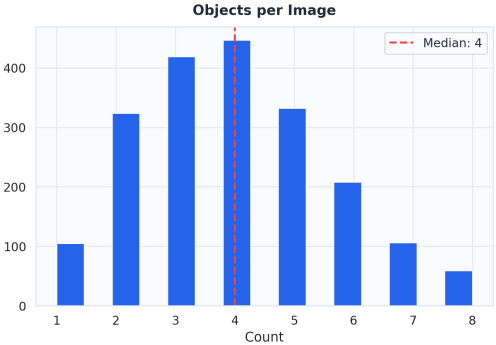
Felipe Taha Sant'Ana

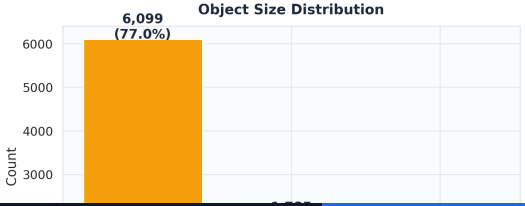
Agenda

1. Dataset
2. mAP Evaluation
3. Speed vs Accuracy
4. Detailed Analysis
5. Conclusions

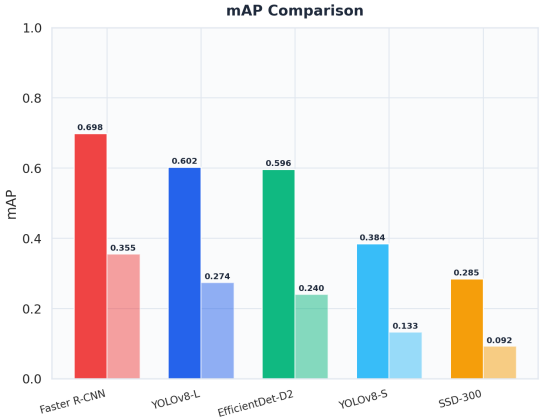
Synthetic Detection Dataset

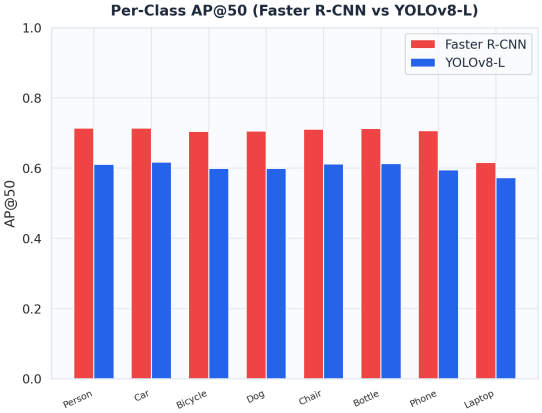






mAP Comparison & Per-Class AP

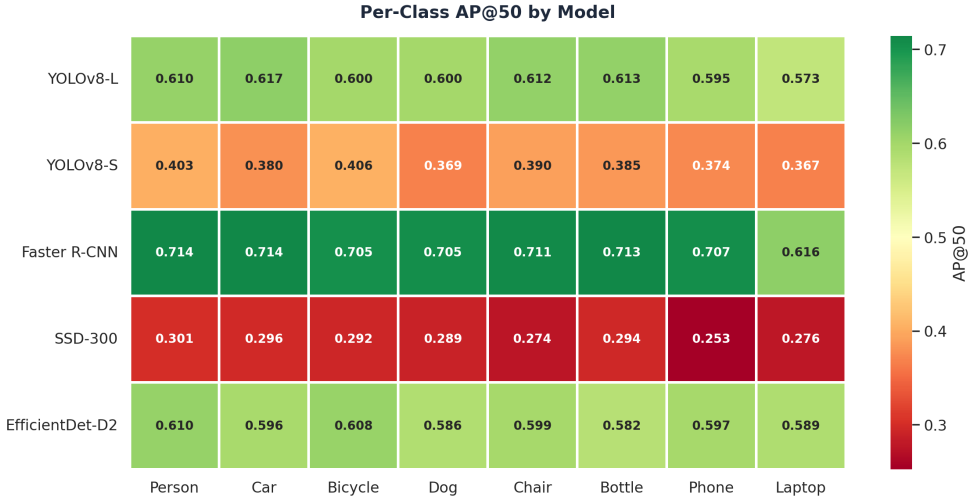




Results

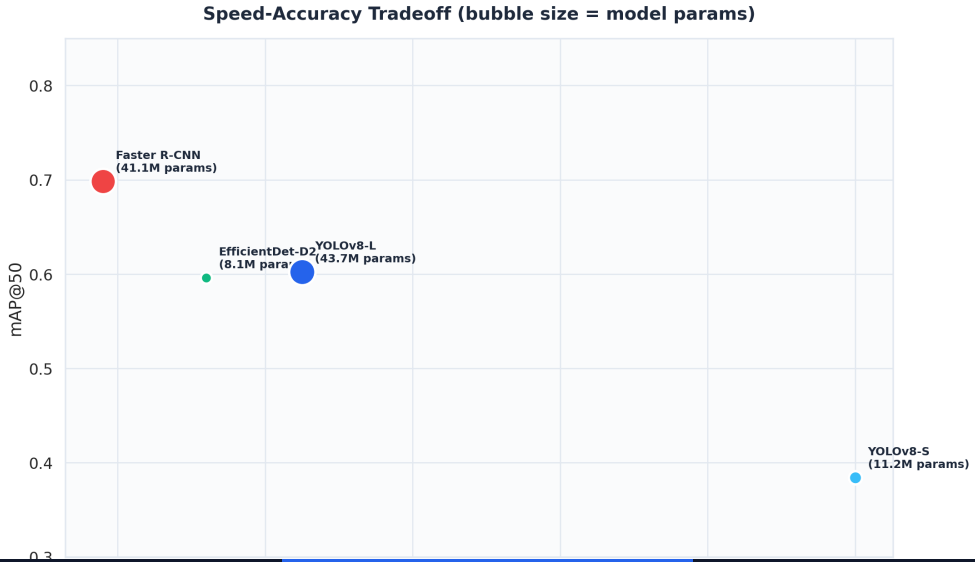
Faster R-CNN leads with mAP@50 = 0.698. YOLOv8-L follows at 0.602 with 2.5x faster inference.

Per-Class AP Heatmap

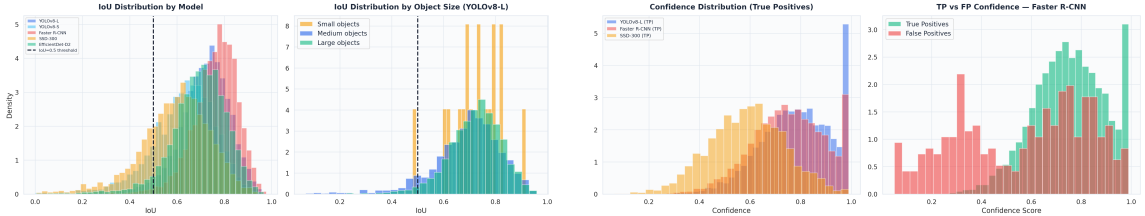


Class-Level Insights

Speed-Accuracy Pareto Frontier



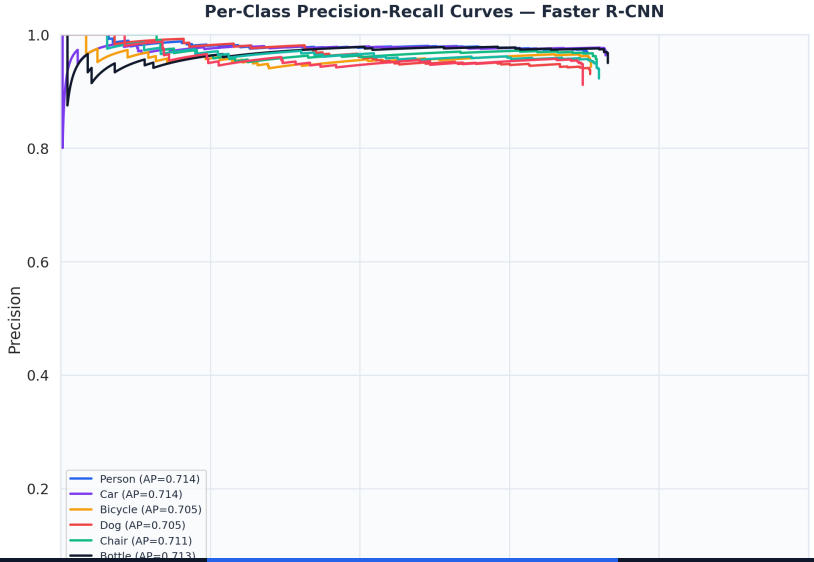
IoU & Confidence Distributions



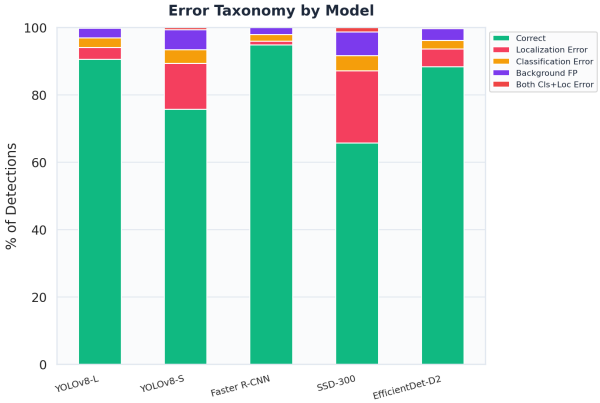
Key Patterns

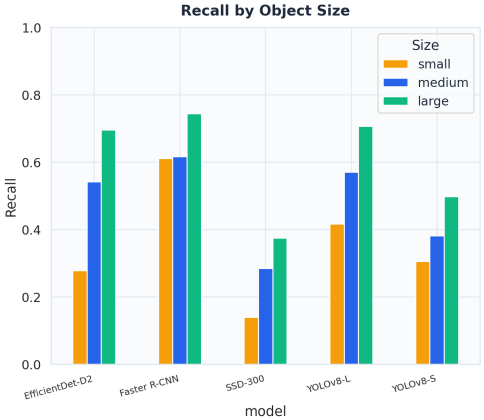
Faster R-CNN produces tighter IoU distributions (better localization). True positive confidence is well-separated from false positive confidence.

Per-Class Precision-Recall Curves



Error Taxonomy & Size Analysis

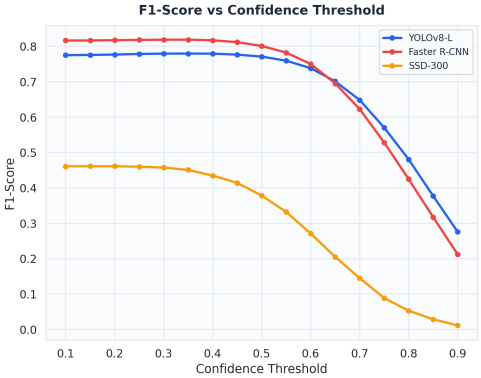


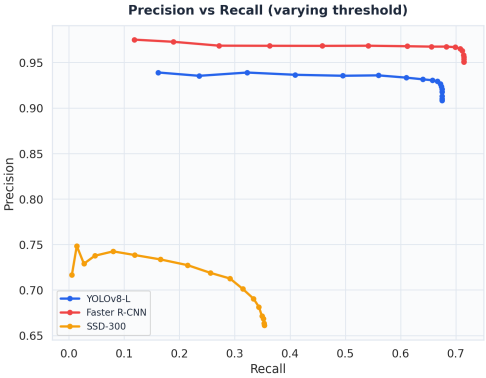


Error Insights

Localization errors dominate over classification errors for all models. Small object recall drops 30–50% compared to large objects.

Confidence Threshold Tuning

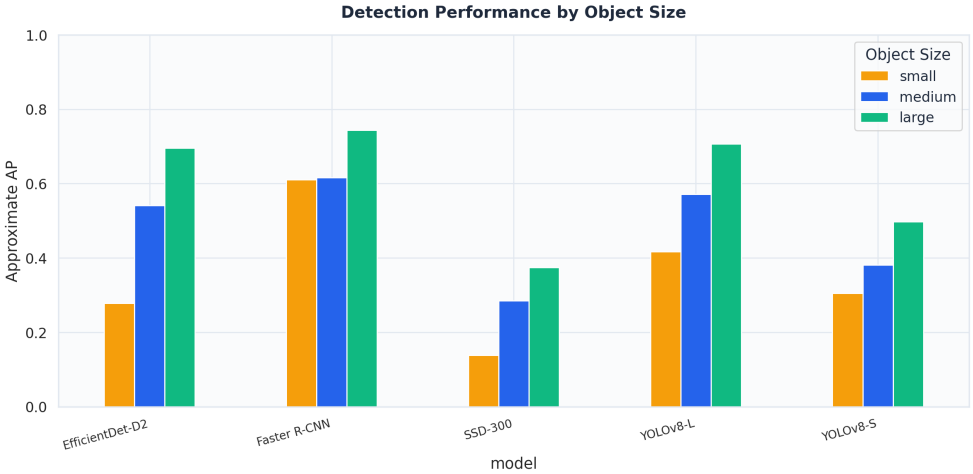




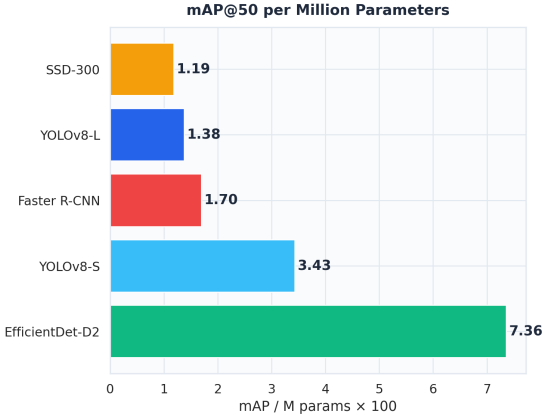
Operating Point Selection

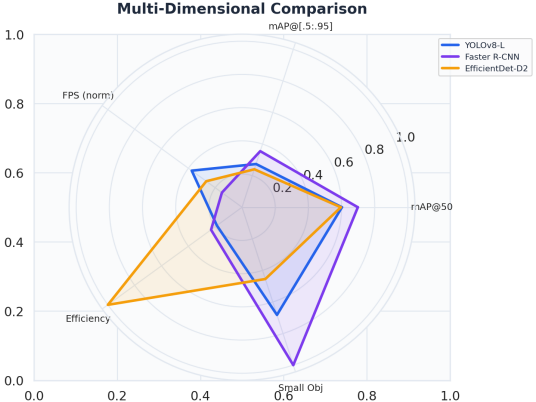
Optimal confidence threshold varies by model. Higher thresholds reduce false positives but miss more objects.

Performance by Object Size



Model Efficiency & Multi-Dimensional Comparison





Executive Summary

Object Detection — Executive Summary

Model Leaderboard

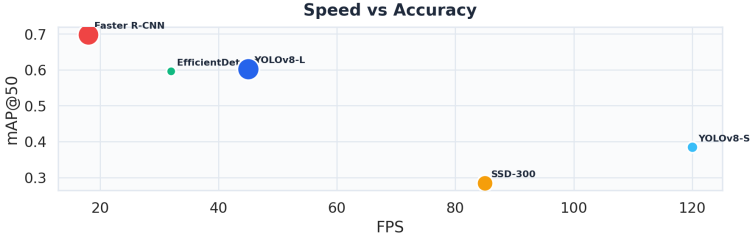
Model	mAP@50	mAP@[.5:.95]	FPS	Params	Efficiency
Faster R-CNN	0.6982	0.3552	18	41.1M	1.70
YOLOv8-L	0.6024	0.2737	45	43.7M	1.38
EfficientDet-D2	0.5959	0.2404	32	8.1M	7.36
YOLOv8-S	0.3841	0.1326	120	11.2M	3.43
SSD-300	0.2846	0.0921	85	24.0M	1.19

```
DETECTION SUMMARY

Dataset:
  Images:        2,000
  Objects:       7,920
  Classes:       8

Best Accuracy:
  Faster R-CNN
  mAP@50:     0.6982
  mAP@[.5:.95]: 0.3552

Fastest:
```



Recommendations & Future Work

Use Case Recommendations

Real-time:	YOLOv8-L (45 FPS, 0.60 mAP)
Edge:	EfficientDet-D2 (8.1M params)
Max accuracy:	Faster R-CNN (0.70 mAP)
Ultra-fast:	YOLOv8-S (120 FPS)

Future Work

1. Transformer detectors (DETR, RT-DETR)
2. Anchor-free architectures
3. Knowledge distillation
4. Data augmentation (Mosaic, MixUp)
5. Domain adaptation
6. 3D detection / video

Thank You

Questions & Discussion

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